Introducing Effective Theories (Wells, Chs. 1–2; Williams) May 20, 2026

Motivating Questions

- What is the EFT "mindset," and how does it differ from the picture of physical theories that has dominated philosophy of science? In what sense are *all* theories effective theories?
- How can we anticipate, parametrize, and even test corrections to a theory *without* possessing the deeper theory of which it is supposed to be a limiting case? What role do symmetries, dimensional analysis, and the identification of natural scales play?
- EFTs are theories valid *within a domain*, with their breakdown built into their mathematical structure. How does this conception of a theory connect to themes the seminar has already developed — Duhem on the symbolic and approximate character of theories, Smith on closing the loop, Wilson on physics avoidance and shifting explanatory landscapes?

Wells on the EFT Mindset

Wells's opening definition (Ch. 1):

> "Effective Theories" are theories because they are able to organize phenomena under an efficient set of principles, and they are effective because it is not impossibly complex to compute outcomes. The only way a theory can be effective is if it is manifestly incomplete. "Everything affects anything" is generally correct, but it saps confidence in our ability to predict outcomes. Effective Theories modify this depressing maxim by pointing out that "most things are irrelevant for all practical purposes." (p. 1)

Three features:

- *Incompleteness is a feature, not a bug*: An EFT systematizes what is irrelevant for the purposes at hand and parametrizes the uncertainty induced by ignoring it. "Correct enough for our purposes in this domain" is the standard, not exact truth.
- *Scale separation*: The "irrelevance of most things" is a substantive empirical claim about the system being modeled—roughly, that physics at one scale is largely insensitive to physics at very different scales. Where scale separation fails (turbulence, critical phenomena), EFTs lose traction.
- *Confronting flaws and revision*: The most useful EFTs are those for which we can characterize the *domain of applicability* and parametrically estimate the size of the corrections we are ignoring. Often there is a controlled procedure for refining the theory by adding further terms.

Galileo's Law as an Effective Theory (Wells, §1.2)

Wells's reconstruction: treat $\ddot{z} = -g$ as a stand-alone discovery, prior to Newton, and ask what an EFT-minded scientist might do with it.

- *Adding the obvious correction*: Symmetry considerations allow a position-dependent correction $\ddot{z} = -g + cz$. Dimensional analysis fixes c to have units of acceleration/length; the only natural acceleration in the problem is g itself, so $c = g/R$ for some length R. The most natural length scale available is the radius of the Earth, $R_e \approx 6400$ km.

- *The constant of tentativeness*: Rather than dogmatically writing $c = g/R_e$, we introduce a dimensionless coefficient η of "order unity," yielding the Adjusted Galilean Law:

$$\ddot{z} = -g\left(1 - \eta\,\frac{z}{R_e}\right) + \cdots$$

  This is *less* predictive than the original GLFB (it has an extra free parameter), but *more* accurate over a wider domain. Newton, of course, would later compute $\eta = 2$.

- *Two morals*: (i) One can anticipate, parametrize, and constrain corrections to a theory *without* possessing the deeper theory. (ii) Writing theories with extra terms of "natural size" consistent with the relevant symmetries is the cornerstone of the EFT approach.

The Harmonic Oscillator Allegory (Wells, Ch. 2)
The harmonic oscillator is ubiquitous because near any equilibrium $V(x) = a_2x^2 + a_3x^3 + a_4x^4 + \cdots$, with the quadratic term dominating for $x < x_{\text{crit}} \sim a_2/a_3$. The chapter traces the response of an imagined community to ever-finer experimental probes of an oscillating "System."

- *Theory 1:* $V(x) = kx^2/2$. One parameter, one observable (the period), trivially fit.
- *Theory 2:* Add a cubic correction $V \supset kx^3/(3\Lambda_A)$ with a new length scale Λ_A. This breaks left-right symmetry and predicts asymmetric "half-periods." Improved measurements show that the half-periods are symmetric to within experimental error. How should the community respond?

  Two strategies: *(i) attribute symmetry to physics*—propose a discrete $x \to -x$ symmetry (or a conformal $x \to \lambda x$ symmetry) that banishes the cubic term; or *(ii) attribute symmetry to suppression*—the cubic term is there, but Λ_A is too large to detect. Wells's warning: fancy symmetry arguments may entrench themselves long after their evidential basis warrants.
- *Theory 3:* Add an even correction $V \supset -kx^4/(4\Lambda_B^2)$; predicts amplitude-dependent period (breaking of scale invariance). Probing the system at higher initial velocity confirms the prediction, fitting $\Lambda_B \approx 95$ m.
- *Theory 4 (a "deep theory" conjecture):* $V = \omega_T^2 L_T[1 - \cos(x/L_T)]$. Numerically indistinguishable from Theory 3 at modest energies, but diverges at large amplitudes. An eventual high-amplitude experiment refutes the conjecture in favor of Theory 3.
- *Limitation of EFT alone:* EFT licenses adding all symmetry-consistent operators, with experiment then constraining their couplings. By itself, it cannot select among arbitrarily many candidate completions. Genuinely deeper insight—when available—orders the operators that may arise.

Pedagogical points: Wells's two chapters present the EFT *mindset* without quantum field theory: scale separation, dimensional estimation of corrections using natural scales, symmetry constraints, the trade-off between completeness and predictivity, and the way precision experiments interact with theory choice. The conceptual moves have analogies in quantum field-theoretic setting that Williams describes, despite changes is the technical machinery. (We will also see that Ruetsche makes use of the example Wells discusses in Chapter 3 of an EFT approach to Newtonian gravity.)

From the EFT Mindset to the Renormalization Group (Williams)
Three central ideas regarding EFTs in quantum field theory:

(1) The autonomy of scales: Williams formulates the precondition for effective theorizing as the requirement that "physical processes at long distances must exhibit minimal dependence on the shorter-distance physics that the effective theory's description leaves out" (p. 3).

(2) The renormalization group: couplings depend on scale: In perturbative QFT one encounters divergent integrals from arbitrarily high momenta. These can be regularized (e.g. by introducing a momentum cutoff Λ) and absorbed into shifts of the bare parameters of the theory, defining *renormalized* couplings that must be extracted from experiment at some scale μ.

The puzzle: the dependence on the arbitrary cutoff Λ has merely been replaced by dependence on the arbitrary measurement scale μ. Asking how the theory defined at scale μ relates to the theory defined at scale μ' leads (with a few lines of algebra) to a differential equation for how the coupling changes with scale—the *renormalization group equation*:

$$\mu \frac{d}{d\mu} g(\mu) = \beta(g(\mu)).$$

The right-hand side is the *beta function.* As Williams notes, "the notion of a 'coupling constant' in QFT is a misnomer: the physical couplings that appear in the renormalized Lagrangian are functions of an energy scale" (p. 17). The fine-structure constant, $\alpha \approx 1/137$ at low energies, takes the value $\alpha \approx 1/128$ at $\mu \sim 80$ GeV.

(3) Wilsonian RG and the effective action: Wilson's picture treats the theory at cutoff Λ as defined only up to that scale; the *Wilsonian effective action* S_W contains, in principle, infinitely many terms consistent with the symmetries:

$$S_W = \int d^4x \left[\frac{1}{2}(\partial\phi_\Lambda)^2 + \sum_{n\geq 0} \lambda_n \, \phi_\Lambda^{2+n} + + \sum_{n\geq 0} \lambda'_n \, (\partial\phi_\Lambda)^2 \phi_\Lambda^n \cdots \right].$$

To pass from a higher cutoff to a lower one, "integrate out" the high-momentum modes: this lowers Λ and modifies the couplings according to their beta functions.

Relevant, marginal, and irrelevant operators. Near a fixed point of the RG flow (a point at which $\beta_i = 0$ for all couplings), interactions are classified by how their couplings behave as the cutoff is lowered:

- *Relevant*: coupling grows at low energies, flowing away from the fixed point;
- *Marginal*: coupling does not change with scale at leading order;
- *Irrelevant*: coupling shrinks at low energies, flowing back into the fixed point.

The terminology echoes Wells's "relevant for the purposes at hand," but with more precision: irrelevant operators contribute to low-energy physics either through inverse powers of Λ (negligible) or through small rescalings of the renormalizable couplings. At the low energies for which a QFT has been empirically tested, only a finite set of interactions—precisely the renormalizable ones—make non-negligible contributions. What had looked like "inexplicable good luck" (the empirical success of the criterion of renormalizability) becomes a structural consequence of RG analysis.

Theory space: The RG equations generate a flow on the space of QFTs coordinatized by the values of the couplings. Theories that flow to the same fixed point form a *universality class.*

Decoupling

The principle that ties the picture together is *decoupling*: given a renormalizable theory of a composite system whose particles have well-separated mass scales, the effects of the heavy physics at scale m_2 on physics at the lower scale m_1 can be absorbed into a renormalization of the parameters of a low-energy theory containing only the light particles. (See also Hartmann's summary of Appelquist–Carazzone (1975).) Decoupling makes the EFT framework powerful: one need not possess the high-energy completion to write down and use a controlled low-energy theory. It suggests a picture of empirical reality as "layered into quasi-autonomous domains" (though we will see qualms about this in Hartmann).

Connections to Seminar Themes

- *Wilson on physics avoidance.* EFT is a disciplined form of "physics avoidance": the irrelevant operators are dropped not by ad hoc fiat but with controlled error bounds. It is the kind of justified separation of scales that Wilson finds missing in many traditional treatments of continua.
- *Pessimistic meta-induction.* If theory change is not abandonment of theoretical commitments but addition of higher-scale physics that modifies parameters of a lower-energy effective theory, the discontinuity threatened by historical theory change looks rather different. (We will return to this with Williams's "Scientific Realism Made Effective.")